SharedPtr

SharedPtr

vector< smacc2::ISmaccState * >

SmaccStatus

map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >

recursive_mutex

atomic< StateMachineInternal Action >

map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >

map< std::string, std ::pair< std::function < std::string()>, boost ::any > >

vector< smacc2_msgs ::msg::SmaccTransitionLogEntry >

SMRunMode

int64_t

SharedPtr

shared_ptr< smacc2 ::introspection::SmaccState Info >

shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >

std::enable_shared _from_this< SmaccStateInfo >

map< const std::type _info *, std::vector < smacc2::introspection ::ClientBehaviorInfoEntry > >

map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccStateReactor Info > > >

map< const std::type _info *, std::vector < std::shared_ptr< smacc2 ::introspection::SmaccEventGenerator Info > > >

int

string

vector< smacc2::introspection ::SmaccTransitionInfo >

vector< std::shared _ptr< smacc2::introspection ::SmaccStateInfo > >

type_info *

#eventsLogPub_
#stateMachinePub

#stateMachineStatus Pub_
#transitionHistory Service_
#transitionLogPub_

#timer_

#currentState_

#status_msg_

#orthogonals_

-eventQueueMutex -m_mutex_

#stateMachineCurrentAction

#stateCallbackConnections

-globalData_

-transitionLogHistory_

-runMode_

-stateSeqCounter_

#rh_

-currentStateInfo_

-parentState_

#stateMachineInfo_

-stateMachine_

+staticBehaviorInfo

+stateReactorsInfo

+eventGeneratorsInfo

+depth_ +stateIndex_

+demangledStateName +fullStateName

+transitions_

+children_

+id_

smacc2::ISmaccStateMachine

+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialized()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectISmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()

smacc2::introspection ::SmaccStateInfo

+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()

shared_ptr< rclcpp ::Logger >

mutex

vector< std::shared _ptr< smacc2::ISmaccClient > >

vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >

vector< std::shared _ptr< smacc2::StateReactor > >

vector< std::shared _ptr< smacc2::SmaccEventGenerator > >

#mutex_ #clients_ clientBehaviors_

#stateMachine_

#stateReactors_

#eventGenerators_

#node_

#stateInfo_

#logger_

smacc2::ISmaccOrthogonal

+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()

smacc2::ISmaccState

+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()

msg< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >

boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >

-mutex_

-statemachine_

-statusPub_

-stateMachine_

-currentOrthogonal_

-signalDetector_

-smaccStateMachine_

-currentState_

#parentState_

smacc2::ISmaccClientBehavior

+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEntry()
# postEvent()
# getCurrentState()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()

publisher_ -mutex_ -subscriber_ -subscriber_ publisher_ #components_ -orthogonal_ #stateMachine_

bool

uint64_t

double

boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >

smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >

atomic< int64_t >

ExecutionModel

vector< smacc2::ISmaccUpdatable * >

vector< std::vector < smacc2::ISmaccUpdatable * > >

SmaccFifoScheduler *

processor_handle

thread

atomic< bool >

publisher_ -mutex_ -initialized_ -refTimestamp_ -refCosLat_ -refLat_ -refLon_ -refSinLat_ #owner_ < void()> -loop_rate_hz_ -lastState_ -executionModel_ -updatableClients_ -updatableStateElements_ -scheduler_ -processorHandle_ -signalDetectorThread_ -end_ -initialized_ -rosInitialized_

smacc2::ISmaccComponent

+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

smacc2::SignalDetector

+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()

cl_px4_mr::MapProjection

+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()

uint8_t

optional< rclcpp::Duration >

optional< rclcpp::Time >

smacc2::SmaccSignal < void()>

float

optional< std::future < int > >

function< void()>

-firstMessage_ -headingGoodForControl -valid_ -velocityValid_ -xyGlobal_ -zGlobal_ -refTimestamp_ -refLat_ -refLon_ -hasPublished_ -goalActive_ projection_ -armingState_ -navState_ -prevArmingState_ -prevNavState_ -xyResetCounter_ -zResetCounter_ -onArmed_ -onDisarmed_ -onLanded_ -onVehicleCrashed_ +onLocalPositionReset +onPositionReceived_ periodDuration_ -lastUpdate_ -heading_ -refLat_ -refLon_ -x_ -y_ -z_ -onFailure_ -onFinished_ -onSuccess_ -isShutdownRequested_ -onEntryThread_ -onExitThread_ -postFailureEventFn -postFinishEventFn -postSuccessEventFn_

cl_px4_mr::CpVehicleLocal Position

+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()

smacc2::SmaccAsyncClient Behavior

+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()

cl_px4_mr::CpVehicleStatus

+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()

cl_px4_mr::CpVehicleCommand

+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()

smacc2::ISmaccUpdatable

+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
+ setUpdatePeriod()
# update()

TrajectorySetpoint

-goalX_ -goalY_ -goalZ_ -xyTolerance_ -zTolerance_ +onGoalReached_

-lastSetpoint_ -localPosition_ -localPosition_ -timeoutMs_ -completed_

cl_px4_mr::CpTrajectorySetpoint

+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()

cl_px4_mr::CpGoalChecker

+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()

-enabled_ -trajectorySetpoint_ #vehicleStatus_ #localPosition_ #vehicleCommand_ -altitude_ -centerX_ -centerY_ -maxRadius_ -spacing_ -speed_ -theta_

cl_px4_mr::CpOffboardKeep Alive

+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()

#trajectorySetpoint_ #goalChecker_

optional< std::chrono ::steady_clock::time _point >

-watchdogStart_

#offboardKeepAlive_

cl_px4_mr::CbPx4ClientBehaviorBase

+ onStateOrthogonalAllocation()
+ ~CbPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()

time_point

-lastUpdateTime_

cl_px4_mr::CbSpiralPattern

+ CbSpiralPattern()
+ onEntry()
+ onExit()
+ update()